

Inter-Subject Correlation of Heart Rate and Attention Modulation

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Abstract

This project focuses on analyzing heart-rate synchrony across individuals to verify if it's modulated by attentional state while watching educational videos. The goal is therefore to replicate a couple of results from Pérez et al. (2021) [1] using data from a newer dataset: the Brain, Body and Behavior Dataset (BBBD) [2].

To achieve this, heart-rate fluctuations were extracted from raw ECG recordings and compared across subjects using ISC-HR (inter-subject correlation of heart-rate).

A hypothesis test with a FDR correction is used to establish statistical significance of the obtained results. ISC-HR is then compared between attentive and distracted conditions through a paired statistical test across all five stimuli.

Hypothesis test and descriptive statistics are employed to analyze synchrony patterns and to detect the effect of attention across stimuli.

The results provide a statistically grounded replication of ISC-HR modulation by attention, discussed in relation to the original paper's finding on which this project is based.

1 Introduction

Heart rate is often treated as a purely autonomic signal, reflecting internal physiological states such as arousal or stress.

A recent study suggests that heart-rate fluctuations can be driven by external stimuli. When multiple individuals are exposed to the same narrative content, their heart rates tend to fluctuate in a similar way.

This phenomenon is known as inter-subject correlation of heart-rate (ISC-HR).

Pérez et al. (2021) [1] proposed that these heart-rate fluctuations are modulated by conscious, attentive processing of the stimulus.

This project aims to replicate two of the main results reported by the analyzed paper.

- ISC-HR is statistically significant across subjects exposed to the same stimulus (R1).
- Synchrony is modulated by attentional state, being higher when subjects are attentive than when they are distracted (R2).

This work uses data from the Brain, Body and Behavior Dataset (BBBD) [2], which is a more recent dataset that includes similar experiments to the original study.

2 Related Work

The original paper [1] computes ISC-HR by correlating each subject’s instantaneous heart-rate fluctuations against the rest of the group, averaging the resulting pairwise correlations in Fisher-Z space, and testing significance via a circular-shift permutation procedure with False Discovery Rate (FDR) correction. The study reports three main findings across four experiments.

The first one (R1) shows that heart-rate fluctuations synchronize across subjects listening to the same auditory narrative, with 17 of 27 subjects having statistically significant ISC-HR.

The second result (R2) shows this synchrony is systematically higher when subjects attend to a stimulus than when they are distracted.

The third result (R3) shows that ISC-HR predicts subsequent memory recall performance for the narrative content, suggesting that the measure reflects conscious engagement rather than an autonomic response to sensory input.

The fourth experiment applies ISC-HR to patients with disorders of consciousness, showing reduced synchrony relative to healthy controls and framing ISC-HR as a potential prognostic marker in a clinical setting.

As already anticipated, this project replicates only R1 and R2 using an experiment equivalent to the original study’s experiment 2.

3 Dataset

This project uses Experiment 2 of the BBBD dataset [2], which contains recordings of ECG, EEG, eye-tracking, and respiration during educational video viewing.

In experiment 2, 31 subjects watch five educational videos, each one viewed twice by the same subject: attentively the first time, distracted by performing a secondary task (silently counting backward) the second time. ECG recordings were sampled at 128 Hz, with stimulus duration ranging from approximately 2.5 to 6.5 minutes depending on the video.

BBBD’s data descriptor [2] explicitly states that the ECG data provided for this experiment are not newly collected, but are the same recordings originally published in Pérez et al., 2021 [1].

ECG preprocessing documentation [2] states that R-wave peaks were detected automatically and then manually corrected. This project instead detects R-peaks directly from the raw ECG signal using an automated procedure, without using BBBD’s provided peak timestamps and without manual review. This choice was made in order to reproduce and understand the original study’s methodology.

Not every subject has a usable recording for every (subject, session, stimulus) combination. Missing files are skipped automatically, so the number of subjects analyzed varies slightly across stimuli and conditions.

4 Methodology

This chapter describes the full process from raw ECG signals to the statistical tests used to replicate R1 and R2.

Most of the process follows the methodology described in the original paper.

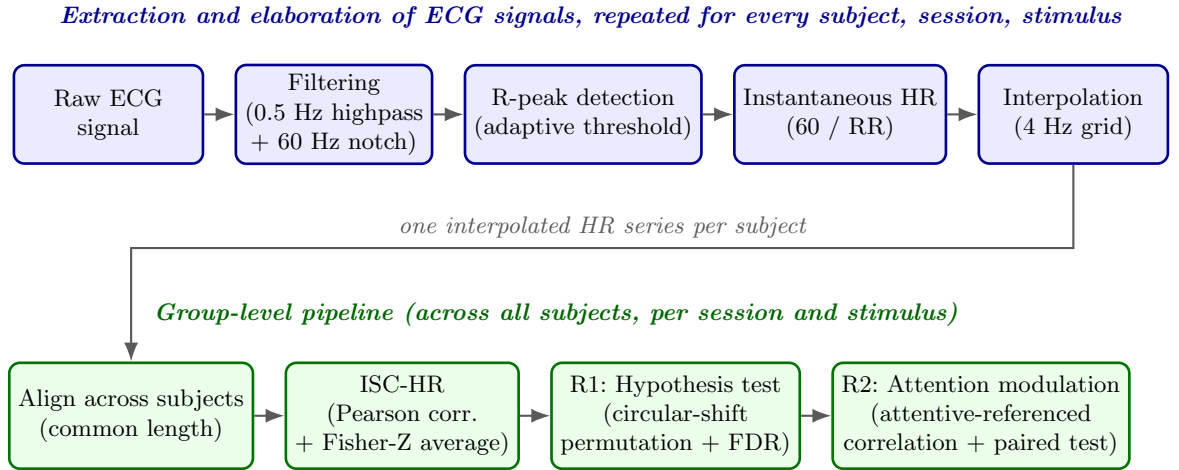


Fig. 1 Full processing pipeline from raw ECG signal to the statistical tests to replicate R1 and R2.

4.1 Signal preprocessing

Raw ECG signals are first high-pass filtered to remove baseline drift, then a notch filter is used to suppress power-line interference. The first high-pass filter used a 0.5 Hz

cutoff, in order to preserve the faster fluctuations corresponding to heartbeats while attenuating baseline drift.

The notch filter targets 60 Hz, which is the frequency of electrical interference from the power supply. Considering the dataset’s 128 Hz sampling rate, the Nyquist limit is therefore 64 Hz. This places 60 Hz close to this edge. Thus the effect of the second filter is marginal.

R-peaks are detected by using an adaptive amplitude threshold. The chosen value is $3 \times$ standard deviations of the filtered signal. This is combined with a 0.3s temporal distance between consecutive detections.

Instantaneous heart rate was computed as the inverse of consecutive R-R intervals $\frac{60}{RR}$ (BPM), where RR is the time interval between two consecutive R-peaks. Then it’s resampled onto a common 4Hz grid via interpolation to allow comparison across subjects.

This last step is made because instantaneous HR series have one value per heart-beat and themselves do not occur at fixed intervals. The requirement for the next step which is computing a Pearson correlation between the HR signals of different subjects requires vectors of equal length, aligned on the same time points.

Subjects whose mean heart rate fell outside a plausible range (40-160 BPM) were automatically excluded from further analysis in order to improve the quality of the results.

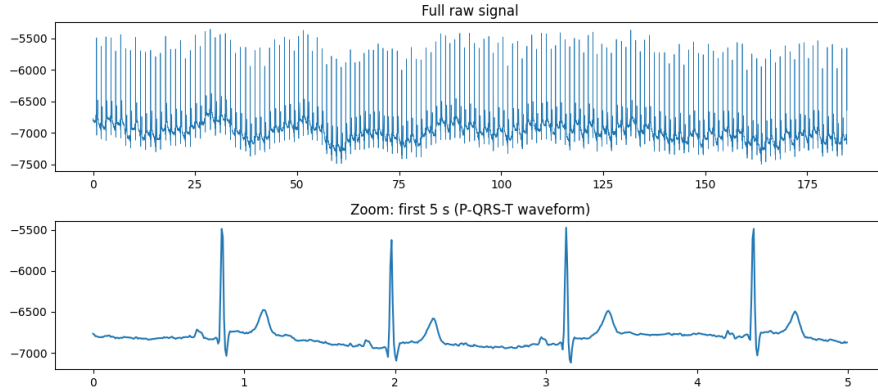


Fig. 2 Full raw ECG signal of a single subject on a single session and one stimulus.

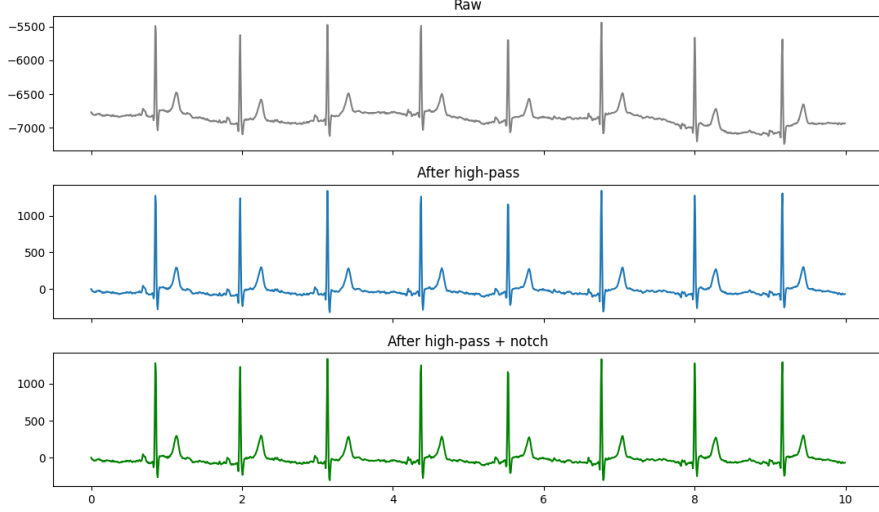


Fig. 3 Raw ECG signal and after filters are applied.

4.2 ISC-HR computation

In order to compute ISC-HR, all subjects' interpolated HR series are truncated to their shortest common length and arranged into a matrix of pairwise Pearson correlations.

Each subject's ISC-HR is computed as follows:

$$\text{ISC-HR}_i = \tanh \left(\frac{1}{N-1} \sum_{j \neq i} \text{arctanh}(r_{ij}) \right) \quad (1)$$

where r_{ij} is the Pearson correlation between subject i and subject j , and self-correlation ($j = i$) is excluded. Correlations are first transformed with the Fisher-Z transform (arctanh) before averaging. The inverse transform ($\tanh$) then maps the average back onto the original correlation scale.

4.3 Hypothesis test

In order to establish statistical significance of each subject's ISC-HR, a circular-shift permutation test is computed.

The subject's HR series is circularly shifted by a random amount and re-correlated against the reference group. This is repeated $B = 10,000$ times to build a null distribution, which is the set of ISC-HR values that subject would obtain purely by chance.

The p-value is the fraction of the B null values that are at least as large as the subject's real observed ISC-HR. It measures how surprising the observed value is under

the assumption of no true synchrony.

A small p-value means it would be rare, by chance alone, to obtain a value this large.

In practice, this fraction is computed as:

$$p = \frac{1 + \text{count}}{B + 1} \quad (2)$$

where count is the number of null values at least as large as the observed ISC-HR. The reason: if the observed value exceeds all B permutations, the plain proportion returns exactly $p = 0$, incorrectly claiming chance could never produce such a result. The $+1$ correction bounds the smallest reportable p-value at $\frac{1}{B+1}$ instead.

Benjamini-Hochberg FDR correction is then applied across all subjects within each combination.

4.4 Attention modulation

Following Pérez et al., 2021 [1] STAR Methods, ISC-HR for the distracted condition was computed against a fixed reference: each distracted subject’s series was correlated against the entire attentive group, rather than against other distracted subjects.

For each of the five stimuli, the attentive-minus-distracted ISC-HR difference was computed per subject (restricted to subjects with a usable recording in both conditions). A Shapiro-Wilk normality test was run once on the pooled differences across all five stimuli to decide between a paired t -test and a Wilcoxon signed-rank test. The chosen test is then applied uniformly to each stimulus individually, testing whether the difference between the attentive and distracted ISC-HR is significantly different from zero.

A significant result ($p < 0.05$) for a given stimulus indicates that attention modulates ISC-HR for that stimulus, addressing R2.

4.5 Implementation choices

Several parameters were not specified in the original paper and required to make implementation choices.

- The amplitude threshold and refractory distance for R-peak detection.
- Physiological HR range used for quality control.
- Interpolation method for resampling.

Two other choices were also made for simplicity:

- Regarding R1, ISC-HR in the distracted condition is computed within-group (distracted subjects correlated against each other) rather than using the attentive-referenced method. The latter is used specifically to replicate R2, where the attentive-distracted comparison requires it.

- This project does not use the original paper’s ANOVA method, which tests both factors (attention $\times$ stimulus) and their interaction together in a single model. R2 instead uses five separate paired tests, one per stimulus [4.4].

5 Results

This section reports the results of R1 (Section 5.1) and R2 (Section 5.2) across all ten (session, stimulus) combinations.

5.1 R1 on all combinations of (session, stimulus)

Figure 4 shows an example correlation matrix, computed across all subjects for the attentive condition (session 1) and stimulus 1.

ISC-HR for each subject is derived from this matrix as described in Section 4.2.

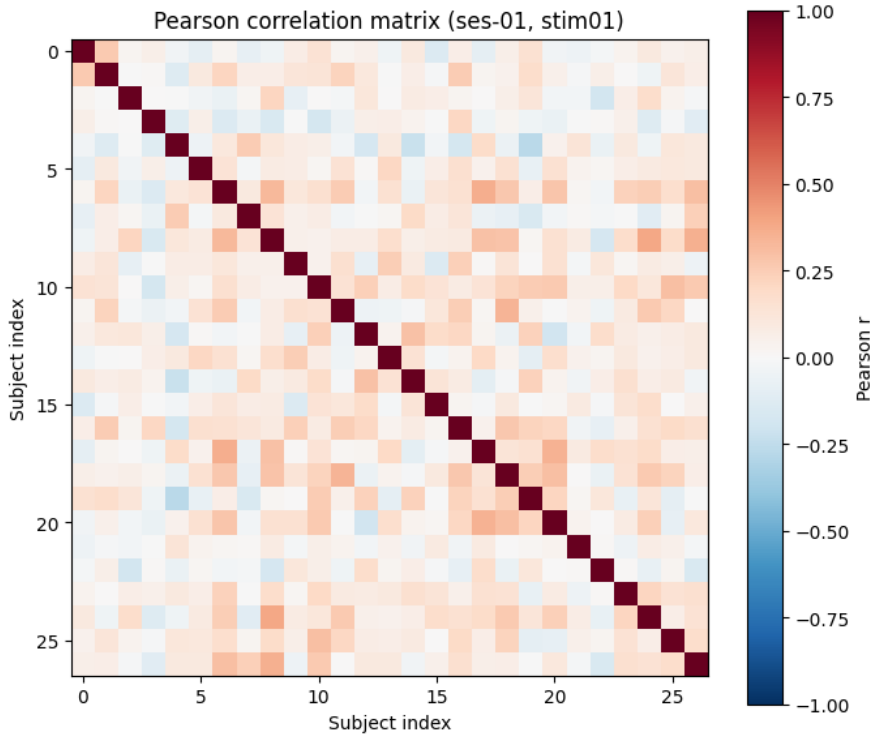


Fig. 4 Pearson correlation matrix for session 1 (attentive), stimulus 1. Entry (i, j) is the Pearson correlation between subject i and subject j . The dark diagonal is each subject with itself ($r = 1$).

Table 1 summarizes ISC-HR results across all five stimuli, for both attentive and distracted (within-group) conditions. In this specific case, as already anticipated in the previous sections, the distracted condition is correlated within-group, not against

the attentive reference.

Stimulus	Condition	Subjects	Mean ISC-HR	Significant (%)
stim01	Attentive	27	0.068	48.1
stim01	Distracted	25	0.002	0.0
stim02	Attentive	27	0.039	18.5
stim02	Distracted	27	-0.002	0.0
stim03	Attentive	28	0.018	17.9
stim03	Distracted	25	0.003	0.0
stim04	Attentive	28	0.032	25.0
stim04	Distracted	25	0.002	0.0
stim05	Attentive	28	0.064	21.4
stim05	Distracted	26	0.008	0.0

Table 1 ISC-HR results across all ten combinations (within-group method for both conditions).

For every stimulus, the attentive condition shows a positive mean ISC-HR with a substantial fraction of subjects individually significant (from 17.9 to 48.1%), well above the $\sim 5\%$ expected by chance alone at $\alpha = 0.05$. The distracted condition, computed within-group, shows essentially no synchrony for any stimulus.

5.2 R2: attention modulation

Table 2 shows, for each stimulus, the mean ISC-HR under both the attentive condition and the attentive-referenced distracted condition, together with the result of the paired comparison between them.

Stimulus	Attentive	Att. sig. (%)	Distracted (ref.)	Dist. sig. (%)	p -value
stim01	0.068	48.1	0.025	4.0	0.0004
stim02	0.039	22.2	0.020	22.2	0.2584
stim03	0.018	17.9	0.008	0.0	0.0802
stim04	0.032	25.0	0.010	0.0	0.0003
stim05	0.064	21.4	0.026	0.0	0.0236

Table 2 Mean ISC-HR and percentage of individually significant subjects (Section 4.3) for the attentive and attentive-referenced distracted conditions, alongside the paired comparison between them, per stimulus. The Wilcoxon signed-rank test is used, chosen over a paired t -test after Shapiro-Wilk rejected normality on the pooled differences at $\alpha = 0.05$ ($W = 0.979$, $p = 0.046$; W close to 1 indicates normality, so this result sits just below the threshold).

Three of the five stimuli (stim01, stim04, stim05) show a statistically significant reduction in ISC-HR under distraction ($p < 0.05$), consistent with R2. The remaining two stimuli do not reach significance, for different reasons discussed below.

stim02 and stim03: different patterns behind similar p -values

stim02 and stim03 both fail to reach significance ($p = 0.258$ and $p = 0.080$ respectively), but for different underlying reasons, visible in Table 2’s individual-significance columns. For stim02, 22.2% of distracted subjects remain individually significant, which is identical to the attentive rate. This is indicating that distraction did not measurably reduce synchrony for this specific stimulus. For stim03, by contrast, 0% of distracted subjects are individually significant, matching the pattern of stim04 and stim05 (which do reach significance); stim03’s weaker result instead reflects its smaller attentive-condition baseline (0.018, the smallest of all five stimuli), which limits how large the attentive-minus-distracted difference can be.

6 Discussion

6.1 Comparison with the original study

Attentive viewing produces significant inter-subject heart-rate synchrony, and this synchrony is reduced under distraction. However, direct numerical comparison with the paper’s reported figures should be treated with some caution.

BBBD’s data descriptor explicitly attributes Experiment 2’s ECG data to Pérez et al., 2021 [1] (Section 3), confirming the same underlying raw recordings are used here. This project detects R-peaks directly from that raw signal, matching the original paper’s automated approach rather than BBBD’s subsequent manual correction, which was apparently introduced only when BBBD re-released the data and it is not used here. However, the paper we based the work on, does not specify the exact detection parameters (amplitude threshold, refractory distance) used in their original `findpeaks` call.

Thus peak agreement with the original study is therefore not guaranteed, even on the same raw signal, since this project’s own parameters (Section 4.5) are an independent choice.

----- *DARIVEDERE* -----

The original paper does not report an individual-subject significance count for Experiment 2 (the design replicated here); it reports only an aggregate attention $\times$ video ANOVA ($F[1, 104] = 70.64$, $p = 6.93 \times 10^{-9}$) and states that its own per-video follow-up paired t -test reaches significance for all five videos individually. As a rough plausibility check only, Experiment 1 — a different, audio-only narrative design with no distraction condition — reports 17 of 27 subjects (63%) individually significant, broadly comparable in order of magnitude to this project’s attentive-condition rates (17.9–48.1%, Table 1), though not a direct like-for-like comparison. This project’s R2 paired comparison reaches significance for three of five stimuli rather than five of five. These differences are plausibly attributable to the peak-detection parameter differences (Section 3), the self-exclusion choice for the reference group (Section 4.4), the QC criterion (Section 4.5), and ordinary sample variability, rather than indicating a failure to replicate the underlying effect.

Additionally, BBBD (2025–2026) postdates [1] (2021) by several years. Although the two publications share authors and BBBD’s data descriptor explicitly ties Experiment 2’s ECG to the original study, this timing difference is a further reason to treat numerical comparisons as checks of plausibility and order of magnitude rather than exact validation.

Confronto con le cifre del paper (con le cautele che abbiamo già scritto), limiti dichiarati (niente ANOVA completa, sensibilità all’interpolazione non testata, scelte QC/esclusione non specificate dal paper), interpretazione del pattern generale.

7 Conclusion

Cosa hai dimostrato, cosa resta aperto (R3, metodo esatto per test multipli, ecc.).

8. References

Paper Pérez et al., documentazione BBBD/Zenodo, eventuali librerie citate.

Declarations

Ai Usage Disclaimer: parts of this project have been realized with the assistance of ChatGPT(model GPT-5.3) and Claude(model Sonnet 4.6) for the following types of tasks:

- Reviewing and refining text during the writing of this report.
- Collecting ideas about results analysis.
- Generation of Python code to support the development.

Every content generated by AI has been reviewed and tested by the author.

References

- [1] P. Pérez, J. Madsen, L. Banellis, B. Türker, F. Raimondo, V. Perlberg, M. Valente, M.-C. Niérat, L. Puybasset, L. Naccache, T. Similowski, D. Cruse, L. C. Parra, and J. D. Sitt. Conscious processing of narrative stimuli synchronizes heart rate between individuals. *Cell Reports*, 36(11):109692, 2021.
- [2] J. Madsen, N. Kuppa, and L. C. Parra. The Brain, Body, and Behavior Dataset (BBBD): Multimodal Recordings during Educational Videos. *Scientific Data*, 13:920, 2026.